



Swiss Institute of
Bioinformatics

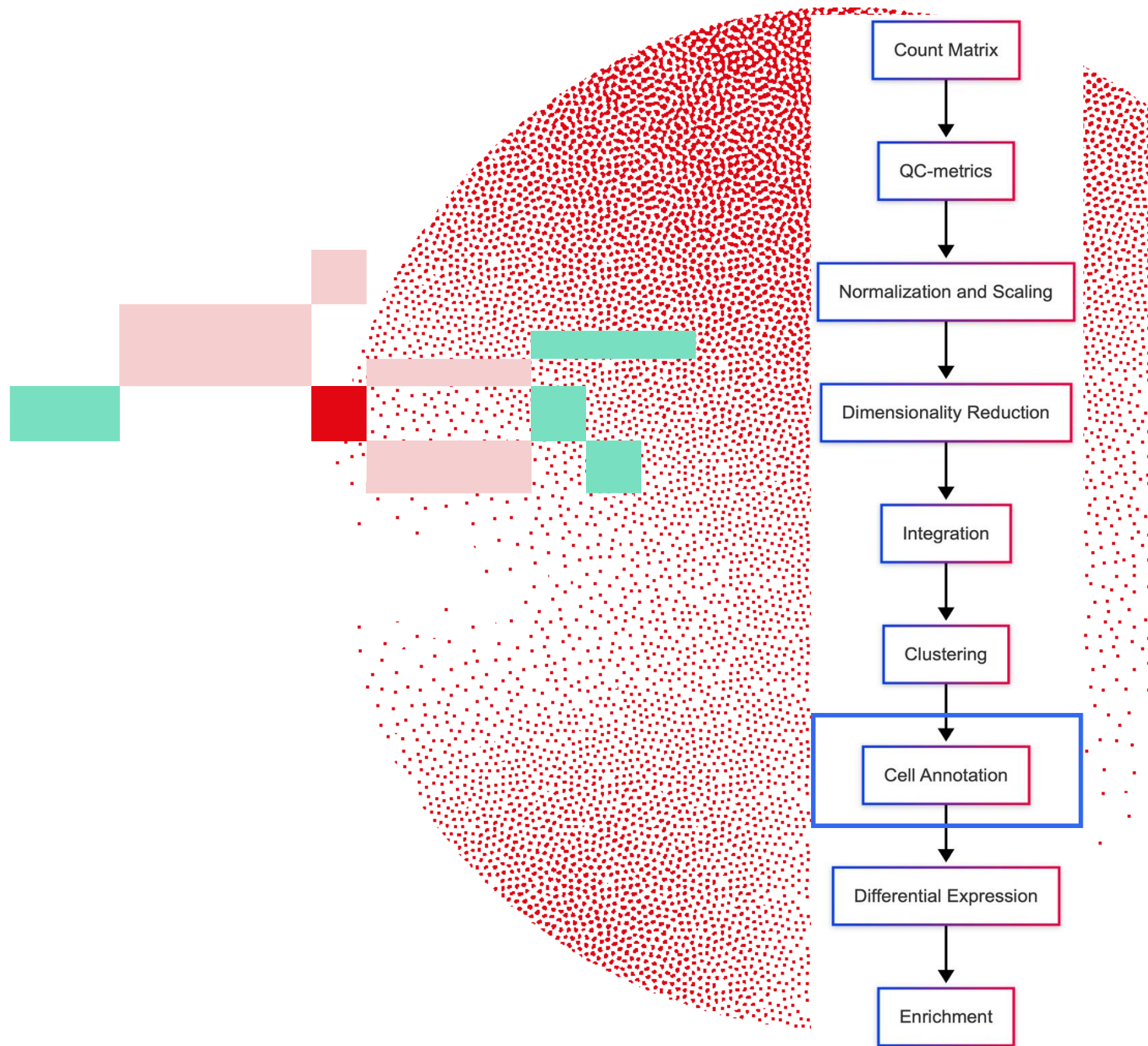
SINGLE-CELL TRANSCRIPTOMICS WITH R

Cell Annotation

Joana Carlevaro Fita

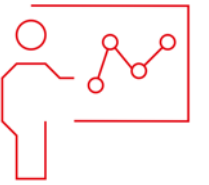
June 16th-18th, 2026

Adapted from previous year courses





Learning objectives



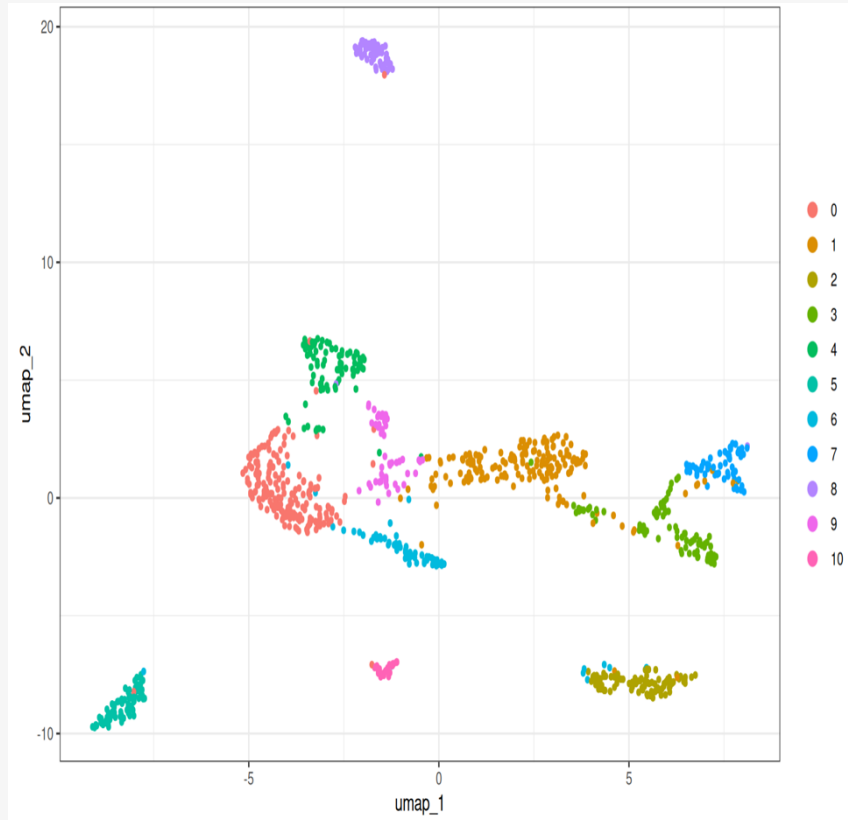
Describe the relevance and challenges of cell annotation

List different cell annotation approaches and applications

Perform cell annotation using singleR package



Cell Annotation



Example of research questions

Identify cell heterogeneity

(different cell abundances)

Identify new cell type subsets

(by function, molecular signature..)

Compare conditions

(ie. tumor vs normal)

Follow cell fate

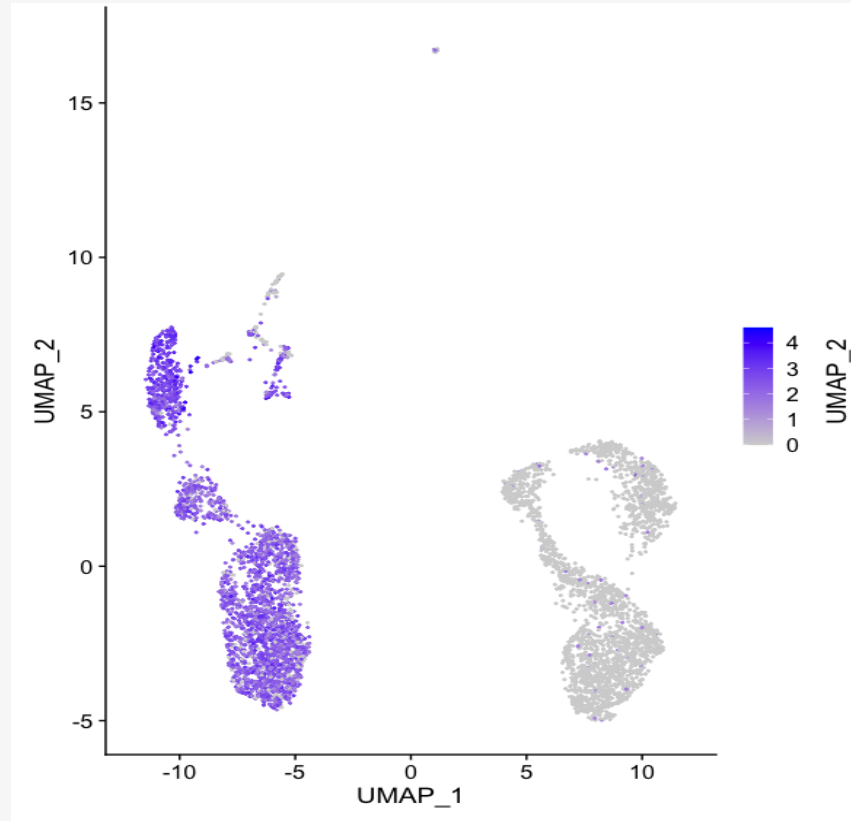
(ie. Development/differentiation)

Annotate cell cycle

(for QC or as biological question)



Cell Annotation



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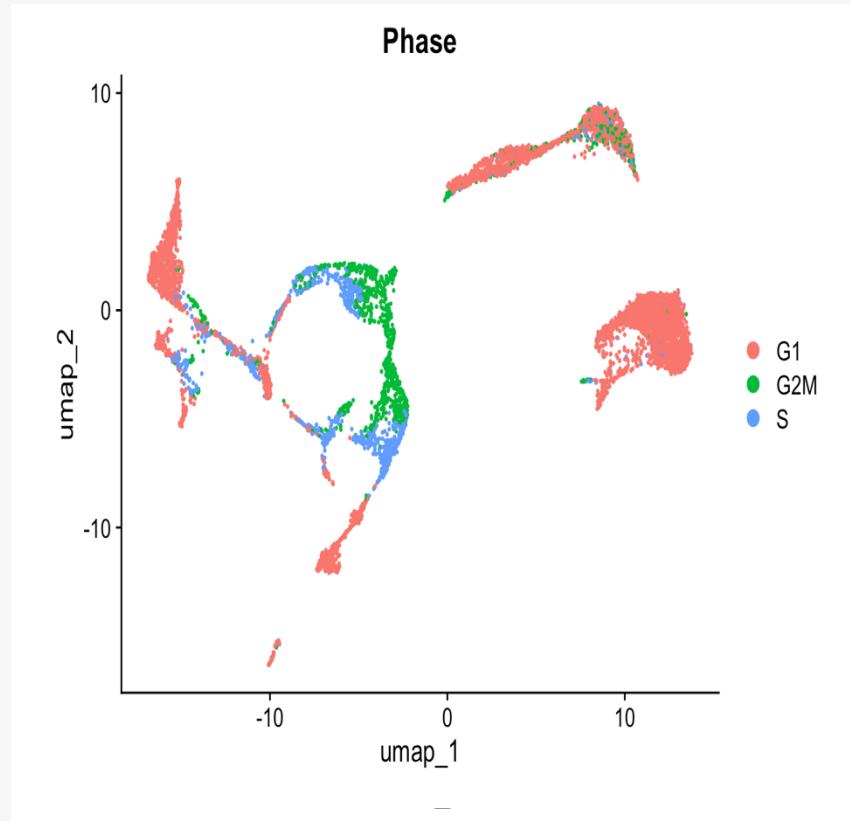
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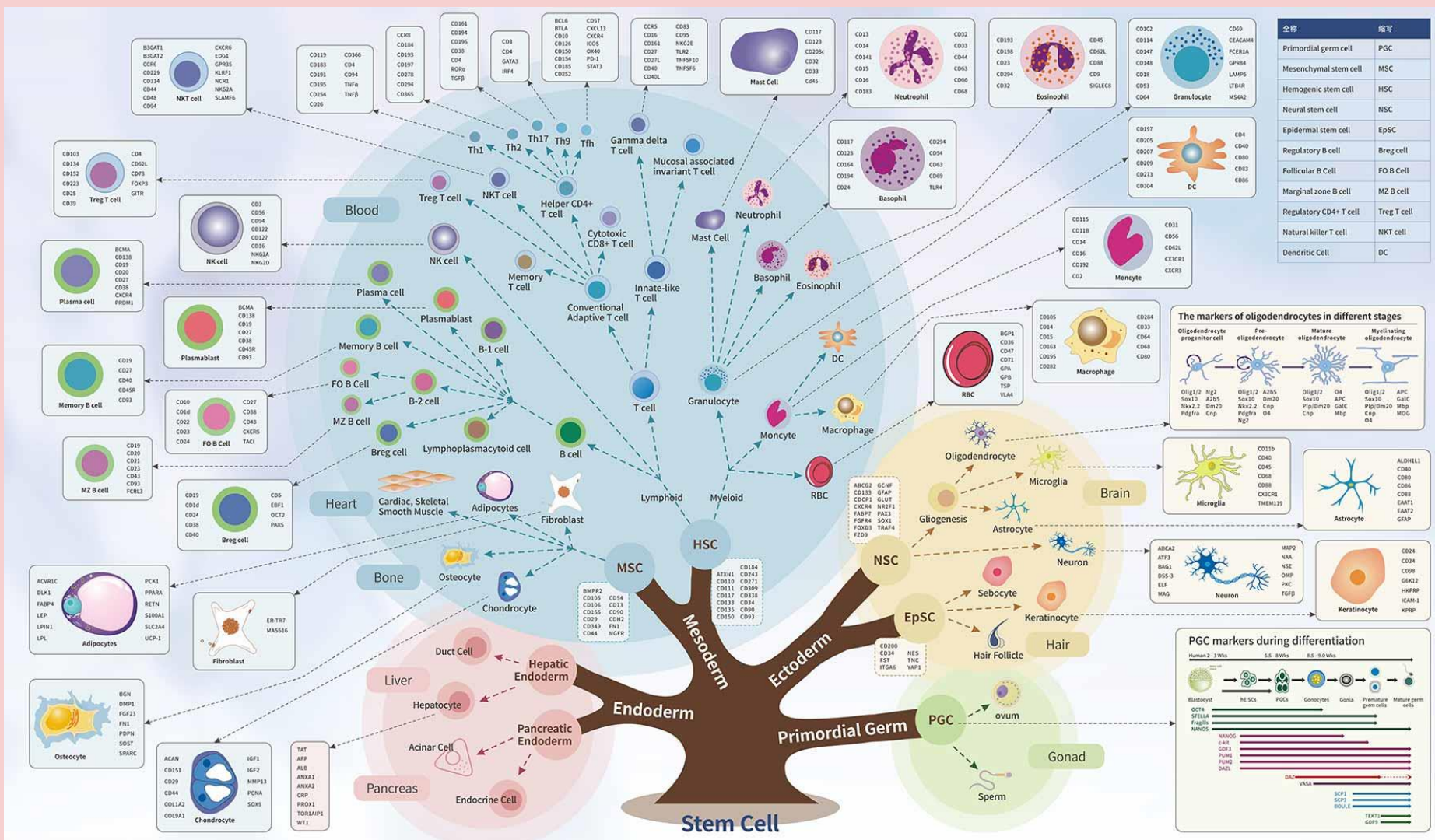
Follow cell fate

(ie. Development/differentiation)

Annotate cell cycle

(for QC or as biological question)

What is a cell type?



Adapted from: <https://www.cusabio.com/statics/images/Cell-Marker.jpg>



Cell Annotation

Cells could be defined in terms of:

- Function
- Location
- Tissue type
- Cell morphology

Identify and label cell clusters
using relevant biological terms

Example of research questions

Identify cell heterogeneity

(different cell abundances)

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(by function, molecular signature..)

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(ie. tumor vs normal)

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Cell Annotation Approaches



Manual



Gene Markers

Identify marker genes from the literature that give us a proxy to explain cell identity



Automatic

Cell Annotation Approaches



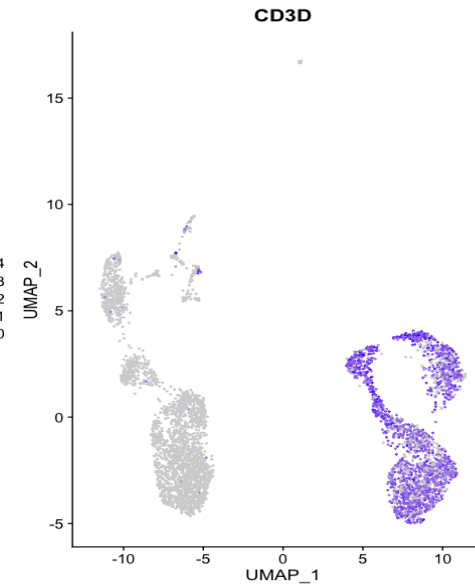
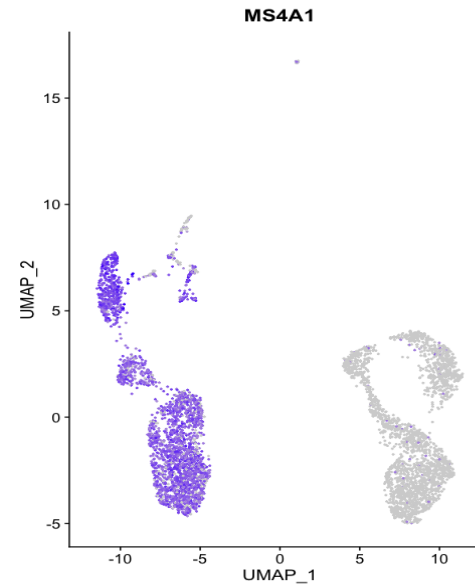
Manual



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Cell Annotation Approaches



Manual

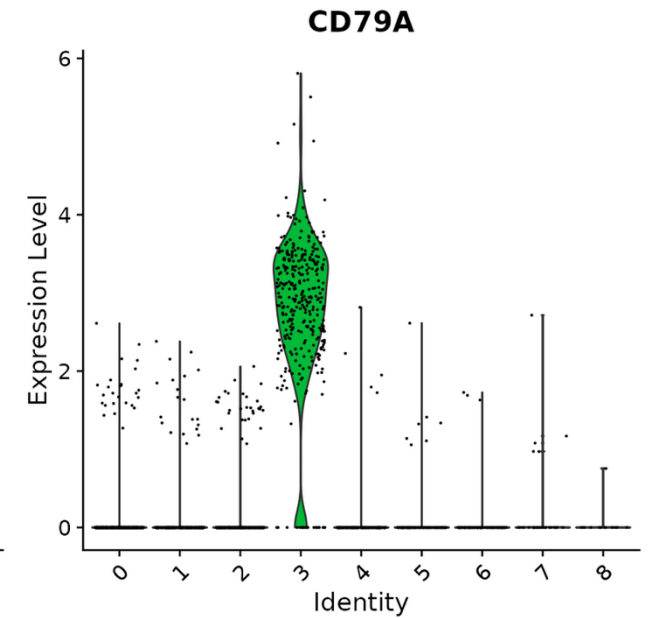
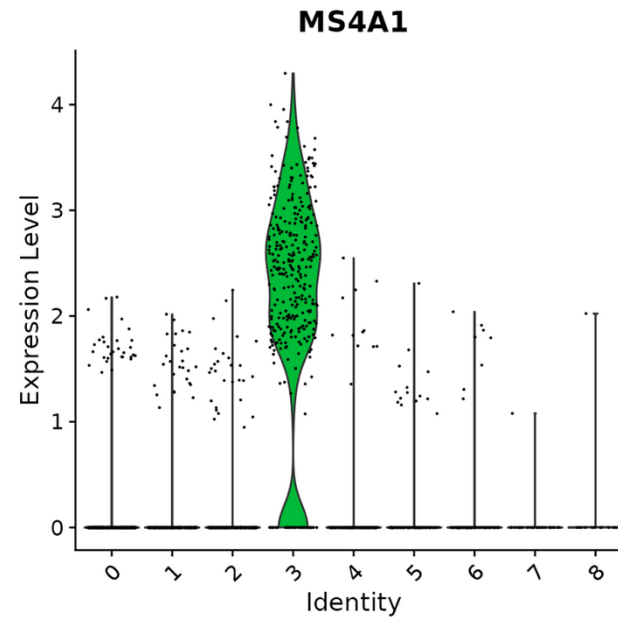


Gene Markers

Identify marker genes from the literature that give us a proxy to explain cell identity



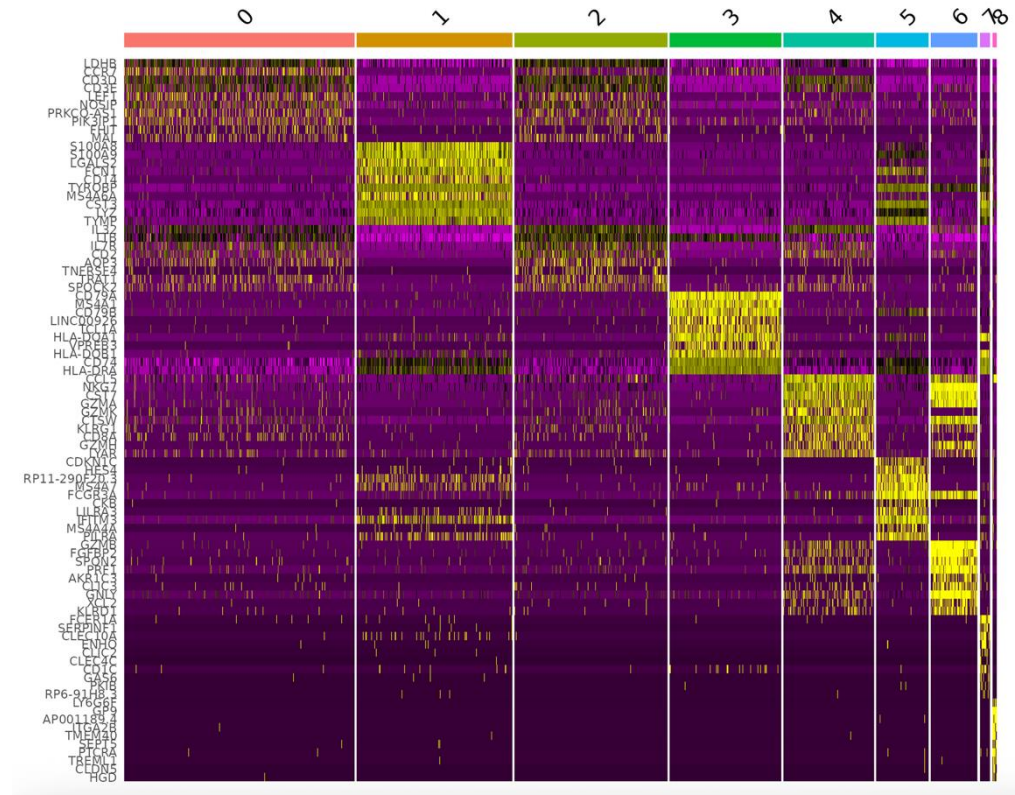
Automatic





Finding differentially expressed features (cluster biomarkers)

Automatic



Cell Annotation Approaches



Manual



Gene Markers



Automatic

It requires expert knowledge

Can be time consuming, subjective

Good markers are not always known or specific to one cell type

Cell surface proteins may not always correlate with mRNA levels

Master transcription factors that drive cell fate are often better markers than cell surface proteins

You may need a combination of genes, one may not be enough

Cell Annotation Approaches

Manual

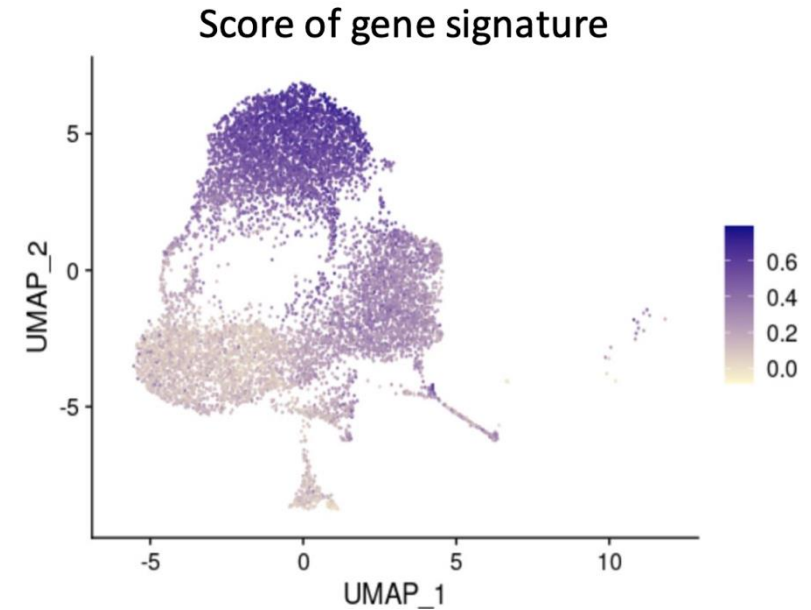


Automatic



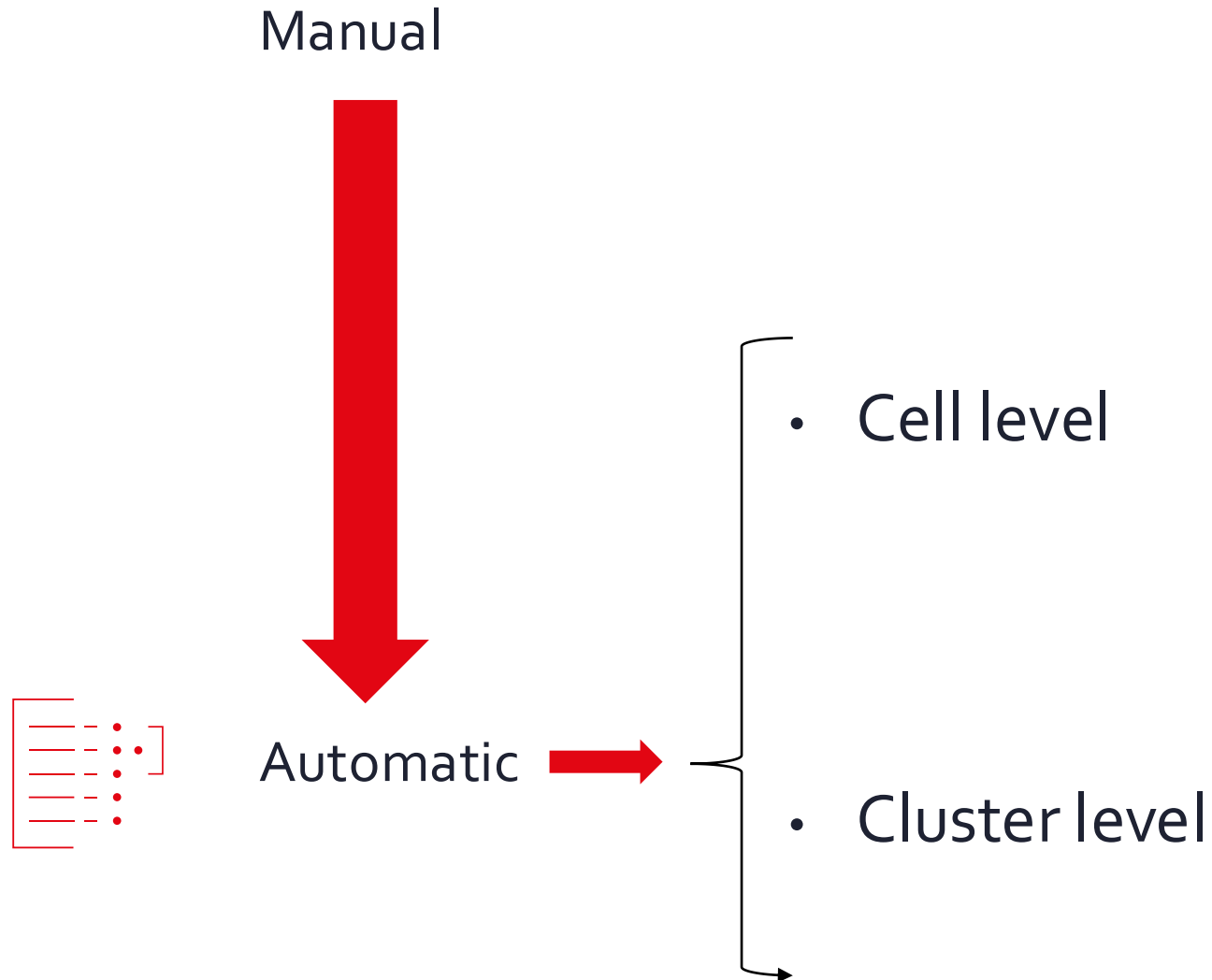
Module Score

Compare expression level of genes belonging to the signature to “control genes” with similar expression level (Tirosh et al 2016, Science 352:6282)



```
Seurat::AddModuleScore(object, features=list() )
```

Cell Annotation Approaches



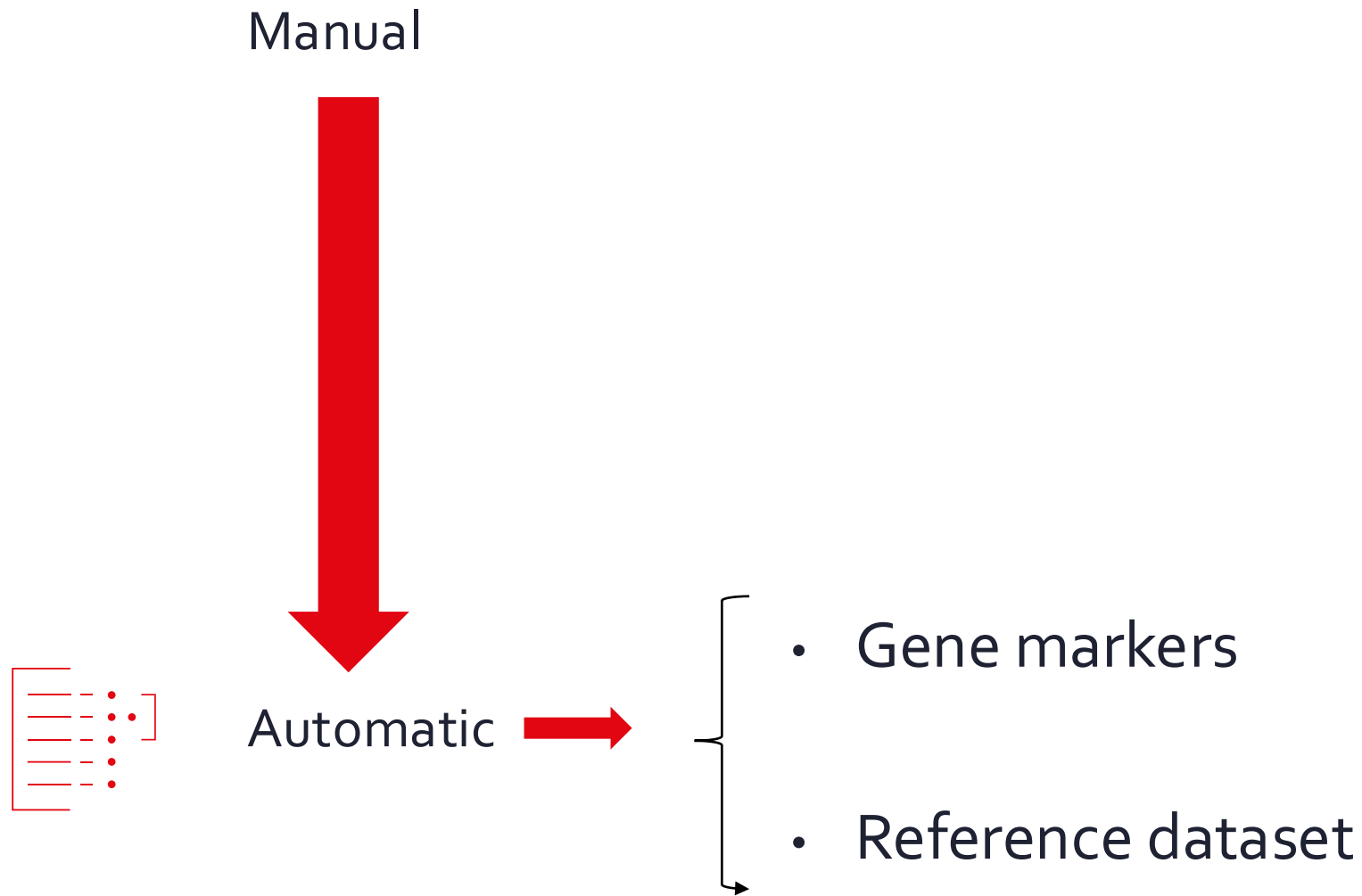
It reduces the chance of missing cell differences / cell types

If read counts are low, more difficult to correctly predict cell type

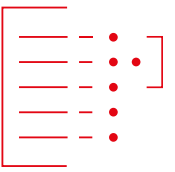
It is faster and could be more accurate (expression levels are based on several cells)

May be challenging for datasets with gradients

Cell Annotation Approaches



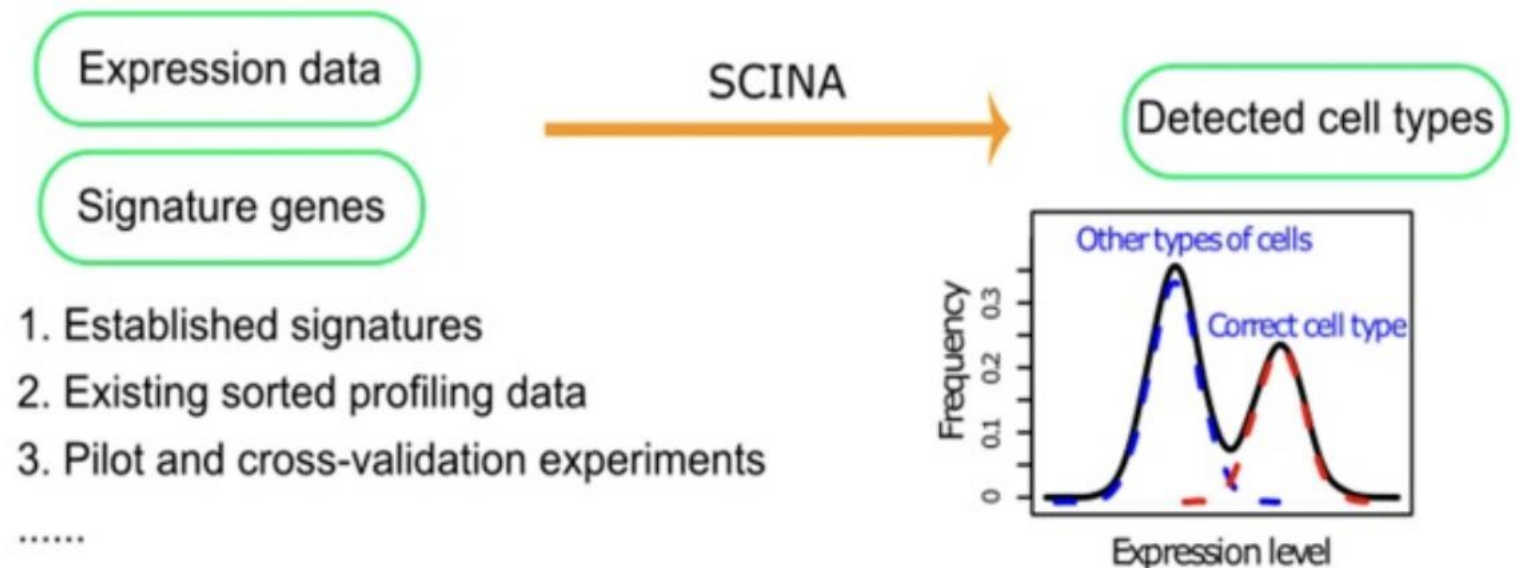
Automatic Annotation



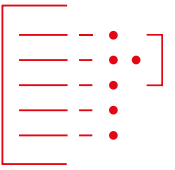
- *Marker based automatic annotation*

Known relationships between gene markers and cell types are obtained from db or literature

- **SCINA**: supervised and automated approach for assigning cell types based on prior knowledge of signature genes and can directly arrive at detected cell types. The gene signatures could come from a variety of sources.



Automatic Annotation

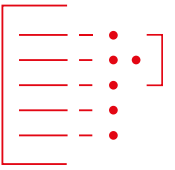


- *Marker based automatic annotation*

Known relationships between gene markers and cell types are obtained from db or literature

- **SCINA**: supervised and automated approach for assigning cell types based on prior knowledge of signature genes and can directly arrive at detected cell types. The gene signatures could come from a variety of sources.
- **AUCell**: uses the “Area Under the Curve” (AUC) to calculate whether a critical subset of the input gene set is enriched within the expressed genes for each cell.
- **GSVA**: Given a db of markers gene set, it identifies enriched sets in within the highly expressed genes in a cluster

Automatic Annotation



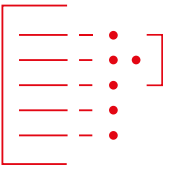
- ***Marker based automatic annotation***

Known relationships between gene markers and cell types are obtained from db or literature

Works well specially if several genes are used

Reliable markers are not available for all cell types

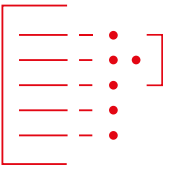
Automatic Annotation



- ***Reference based automatic annotation***

ScRNAseq data to be annotated (query) is compared to a similar and expertly annotated reference dataset (reference). Reference label transferred to query

Automatic Annotation

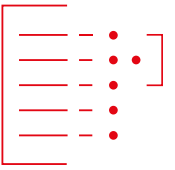


- **Reference based automatic annotation**

ScRNAseq data to be annotated (query) is compared to a similar and expertly annotated reference dataset (reference). Reference label transferred to query

- **Scmap**: *one of the best performing tools. High accuracy and good avoidance of incorrect labelling when there are novel cell types*
- **SingleCellNet**: *Good accuracy if all cell types are present, low if data is incomplete or is a poor match*
- **SingleR**: *based on spearman rank correlation. An advantage is that it includes a general database for reference datasets.*

Automatic Annotation



- ***Reference based automatic annotation***

ScRNAseq data to be annotated (query) is compared to a similar and expertly annotated reference dataset (reference). Reference label transferred to query

Relies on available datasets and resources may be incomplete.

Depends on quality of reference dataset

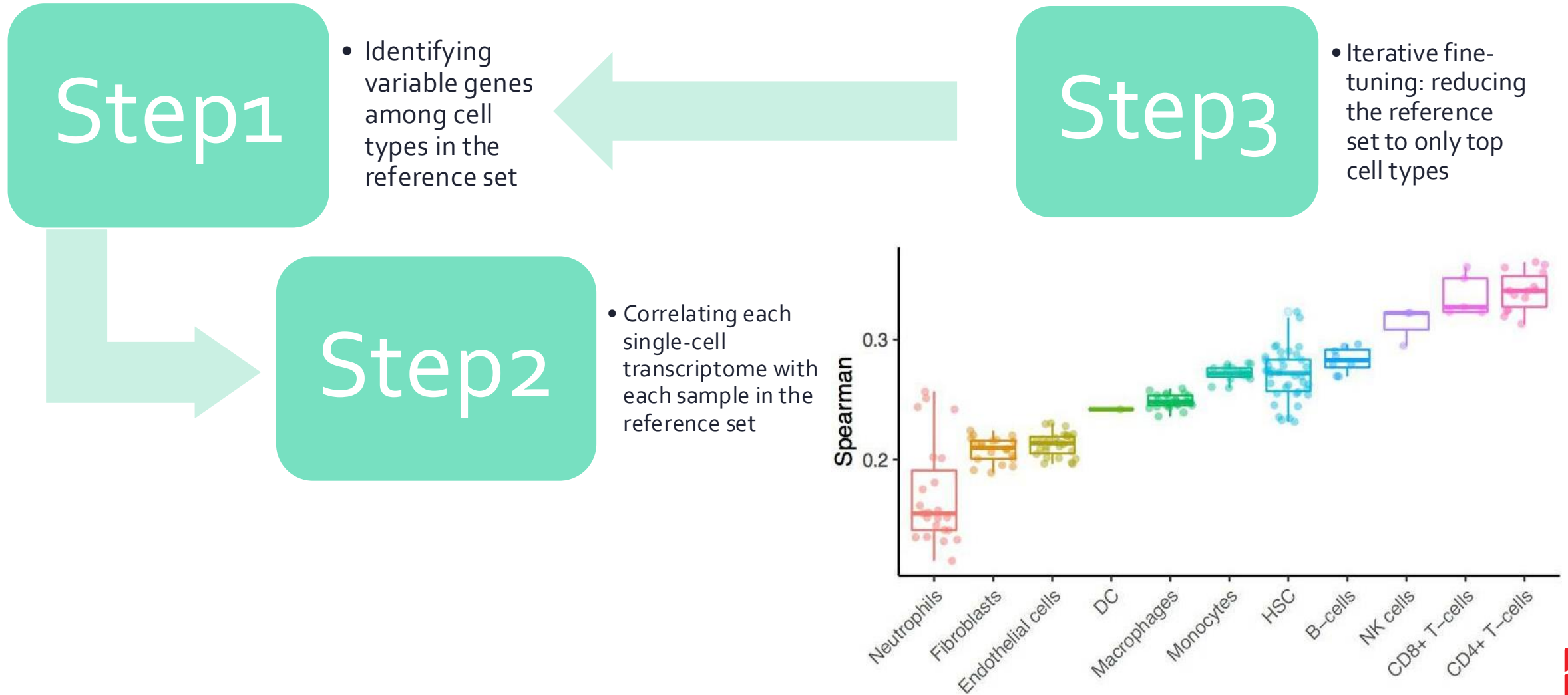
It needs further curation

Automatic annotation



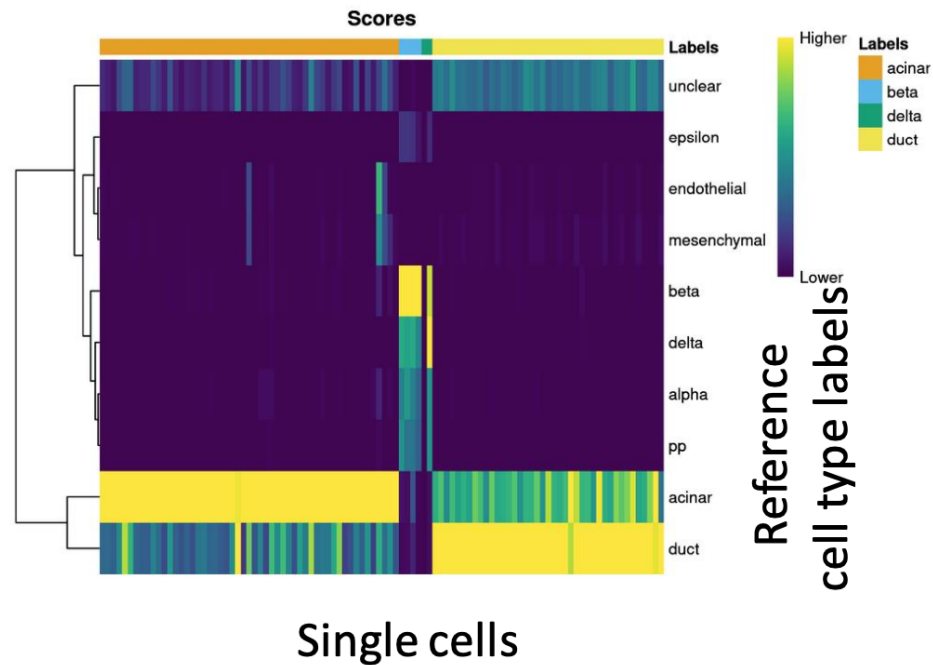
- Easy access to rich reference data:
 - HPCA: hand-annotated Human Primary Cell Atlas 37 main types, subtypes, 713 samples
 - BluePrint +ENCODE
24 main types, 43 subtypes, 259 bulk RNAseq samples
 - Mouse: ImmGen and 'mouse.rnaseq' (brain-specific)
- Classifies cells to both main types and subtypes, performs both single cell-wise and cluster-wise annotation
- SingleR can also be used to evaluate similarity to a custom reference

SingleR - Steps

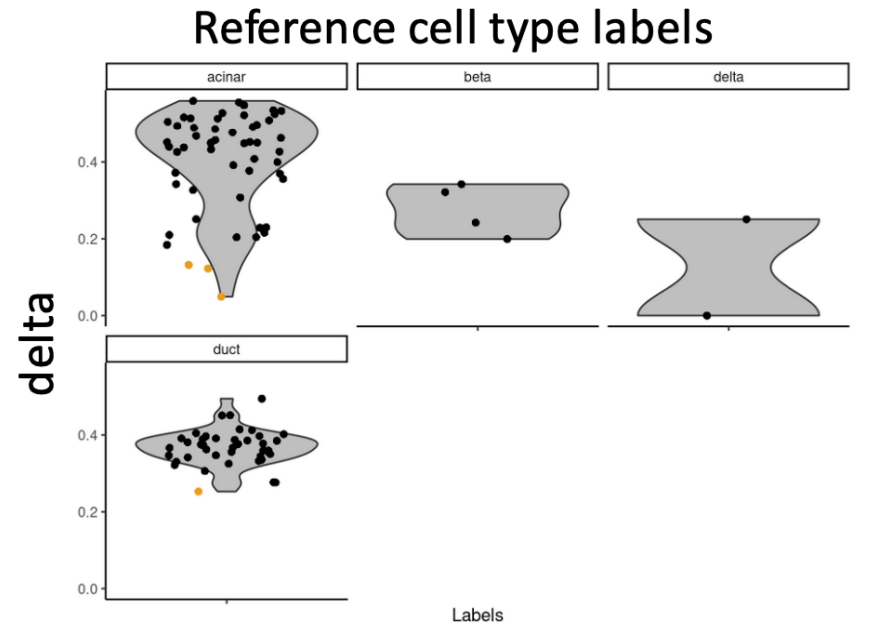


SingleR – annotation diagnostics

Heatmap of scores per cell for each label:



```
plotScoreHeatmap(singleR.object)
```



“delta”: the difference between the score for the assigned label and the median across all labels for each cell

```
plotDeltaDistribution(singleR.object)
```

Try them during the practical exercise

Methods available

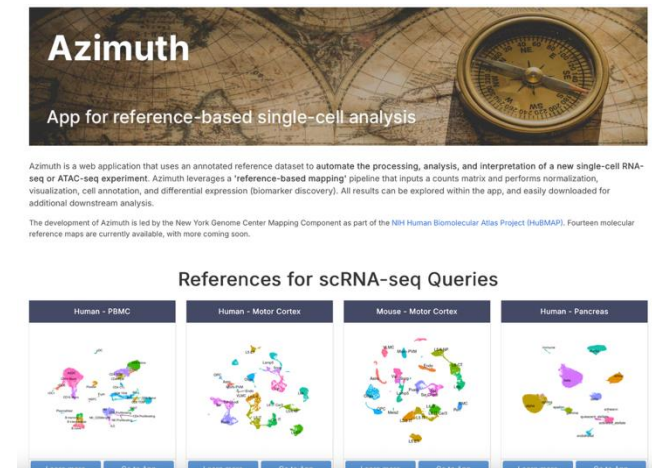
Tool	Type	Language	Resolution	Approach	Allows 'None'	Notes
singleCell Net ⁴²	Reference based	R	Single cells	Relative-expression gene pairs + random forest	Yes, but rarely does so even when it should ³³	10-100× slower than other methods; high accuracy
scmap-cluster ⁴¹	Reference based	R	Single cells	Consistent correlations	Yes	Fastest method available; balances false-positives and false-negatives; includes web interface for use with a large pre-built reference or custom reference set
scmap-cell ⁴¹	Reference based	R	Single cells	Approximate nearest neighbors	Yes	Assigns individual cells to nearest neighbor cells in reference; allows mapping of cell trajectories; fast and scalable
singleR ⁴³	Reference based	R	Single cells	Hierarchical clustering and Spearman correlations	No	Includes a large marker reference; does not scale to data sets of $\geq 10,000$ cells; includes web interface with marker database
Scikit-learn ¹⁰²	Reference based	Python	Multiple possible	k-nearest neighbors, support vector machine, random forest, nearest mean classifier and linear discriminant analysis	(Optional)	Expertise required for correct design and appropriate training of classifier while avoiding overtraining
AUCell ¹⁰³	Marker based	R	Single cells	Area under the curve to estimate marker gene set enrichment	Yes	Because of low detection rates at the level of single cells, it requires many markers for every cell type
SCINA ³⁴	Marker based	R	Single cells	Expectation maximization, Gaussian mixture model	(Optional)	Simultaneously clusters and annotates cells; robust to the inclusion of incorrect marker genes
GSEA/GSVA ^{36,104}	Marker based	R/Java	Clusters of cells	Enrichment test	Yes	Marker gene lists must be reformatted in GMT format. Markers must all be differentially expressed in the same direction in the cluster
Harmony ¹⁰⁵	Integration (Box 2)	R	Single cells	Iterative clustering and adjustment	Yes	Integrates only lower-dimensional projection of the data; seamlessly integrated into Seurat pipeline; may overcorrect data
Seurat-canonical correlation analysis ¹⁰⁶	Integration (Box 2)	R	Single cells	MNN anchors + canonical correlation analysis	Yes	Accuracy depends on the accuracy of MNN anchors, which are automatically-identified corresponding cells across data sets
mnnCorrect ¹⁰⁷	Integration (Box 2)	R	Single cells	MNN pairs + singular value decomposition	Yes	Accuracy depends on the accuracy of MNN pairs (cells matched between data sets). Referred to in Box 2
Linked inference of genomic experimental relationships (LIGER) ¹⁰⁸	Integration (Box 2)	R	Single cells	Non-negative matrix factorization	Yes	Allows interpretation of data set-specific and shared factors of variation. Referred to in Box 2

Databases with cell type markers genes

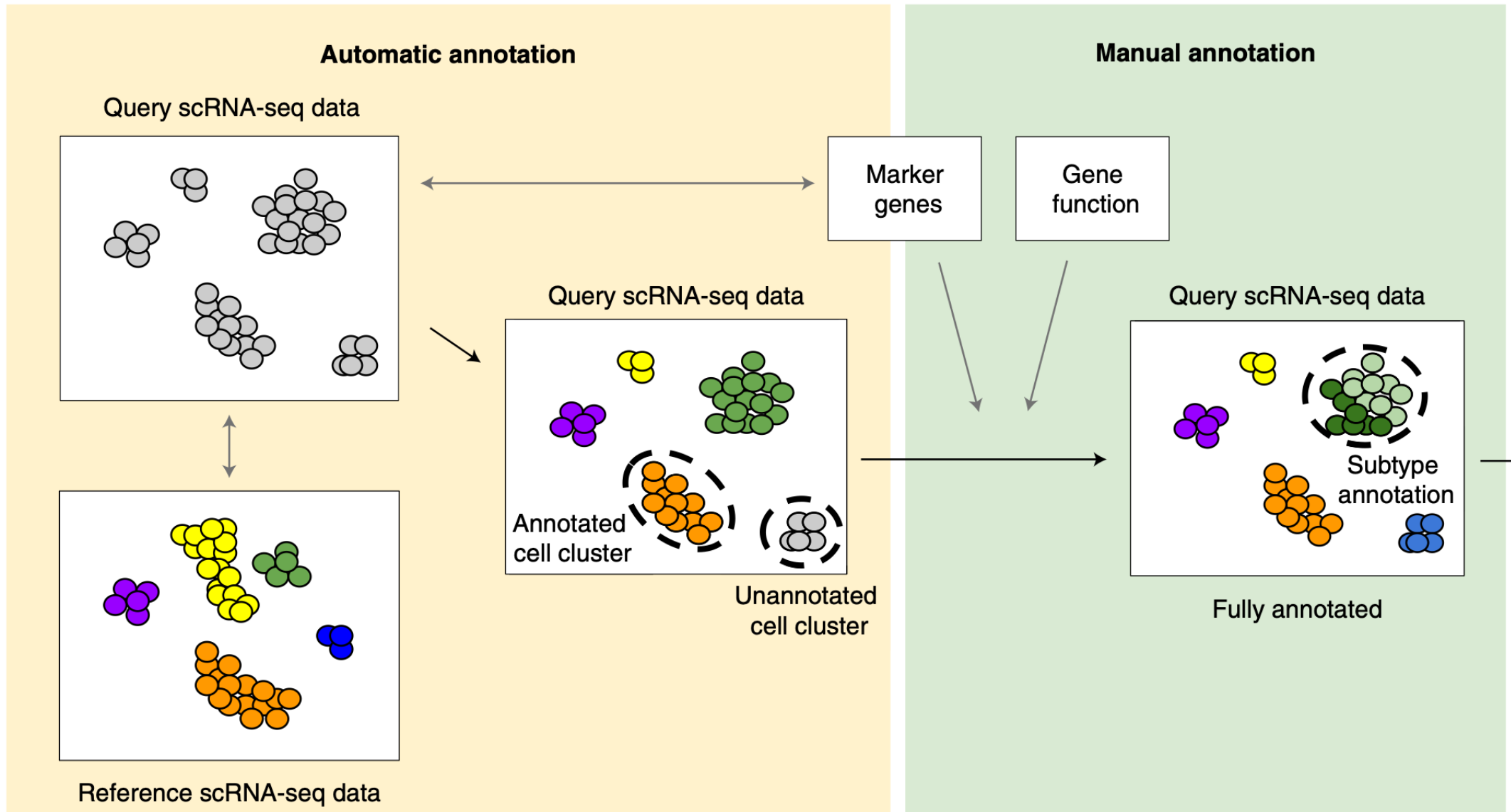
- PanglaoDB <https://panglaodb.se/> (mouse and human)
R: <https://cran.r-project.org/web/packages/rPanglaoDB/index.html>
- CellMarker (mouse and human): <http://bio-bigdata.hrbmu.edu.cn/CellMarker/>
- SingleR <https://github.com/dviraran/SingleR> (Aran et al. 2019), access via celltex package, e.g. human primary cell atlas (microarrays, bulk)
- Human Cell Atlas <https://www.humancellatlas.org> (Regev et al) single cell RNA seq atlas, also some mouse data
- Single cell portal: https://singlecell.broadinstitute.org/single_cell

Reference datasets available

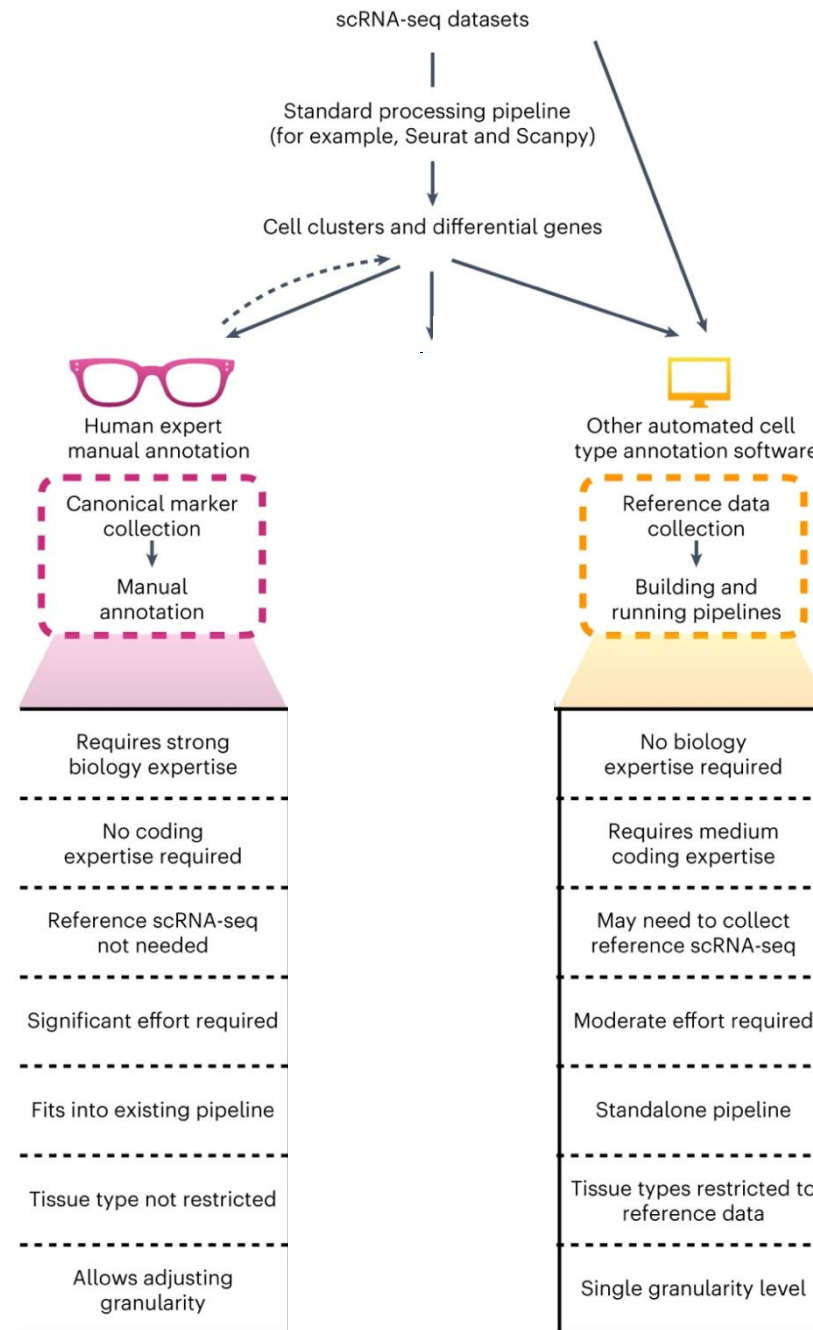
- Single cell portal: https://singlecell.broadinstitute.org/single_cell (10 species)
- Single Cell Expression Atlas: <https://www.ebi.ac.uk/gxa/sc/experiments/E-MTAB-10662/results/cell-plots> (single cell expression across species)
- Gene Expression Omnibus (GEO)
- Azimuth <https://azimuth.hubmapconsortium.org/> (app for reference-based single-cell analysis)



Summary



Summary



Try different methods

! Needs manual curation

Depends on quality and completeness of reference

